

1. Label the animal cell diagram below. Choose from: nucleus, mitochondria, cell membrane, cytoplasm, ribosome, vacuole, lysosome.

2. Complete the plant vs animal cell comparison from memory:

3. Microscope calculations — show all working:

4. Imagine you are looking at a cell under a microscope and you can see: a nucleus, mitochondria, cell membrane, and a very large central vacuole full of liquid. Is this an animal cell or a plant cell? Explain ALL the evidence. Write at least 4 sentences.

5. The lesson says the Fundamental Principles of Cell Theory took nearly 200 years to establish after Hooke first saw cells. Suggest TWO reasons why it might take so long for

a scientific theory to be formally established, even after the first observations are made.

Reason 1: _____

Reason 2: _____

6. 'Structure and function are linked in cells.' Using mitochondria AND chloroplasts as examples, explain this statement. Why does a palisade mesophyll cell (in a leaf) have many chloroplasts? Why does a muscle cell have many mitochondria?
